DATA STRUCTURES

Stack

A Linear Data Structure Following LIFO

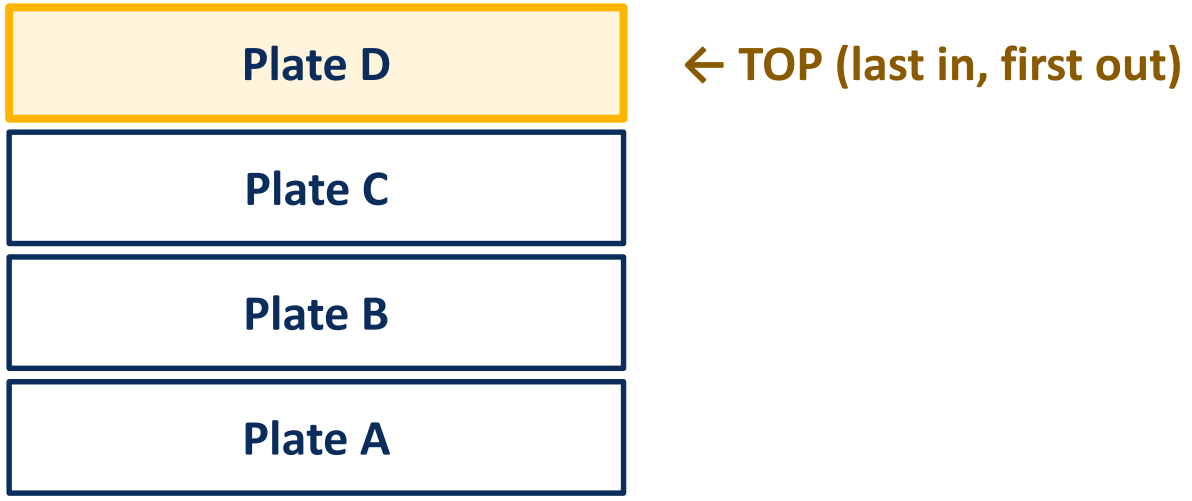
Push & Pop

Real-World Uses

Array Implementation

What is a Stack?

A Stack is a linear data structure that follows LIFO — Last In, First Out. The last element added is the first one removed.



Real-Life Analogies

A Stack of Plates

You always add a new plate on top, and you always take the top plate off first — never from the bottom or middle.

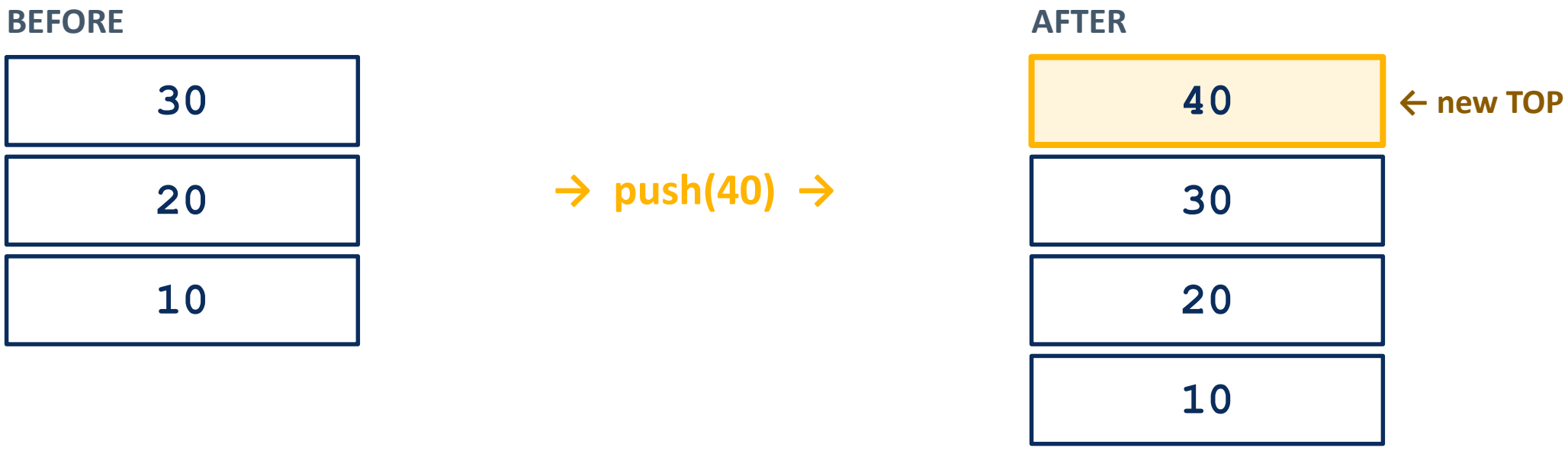
Browser Back Button

Each page you visit is "pushed" on top. Pressing Back "pops" you to the last page you were on — the most recent one first.

Ask students: what other examples follow this last-in-first-out pattern?

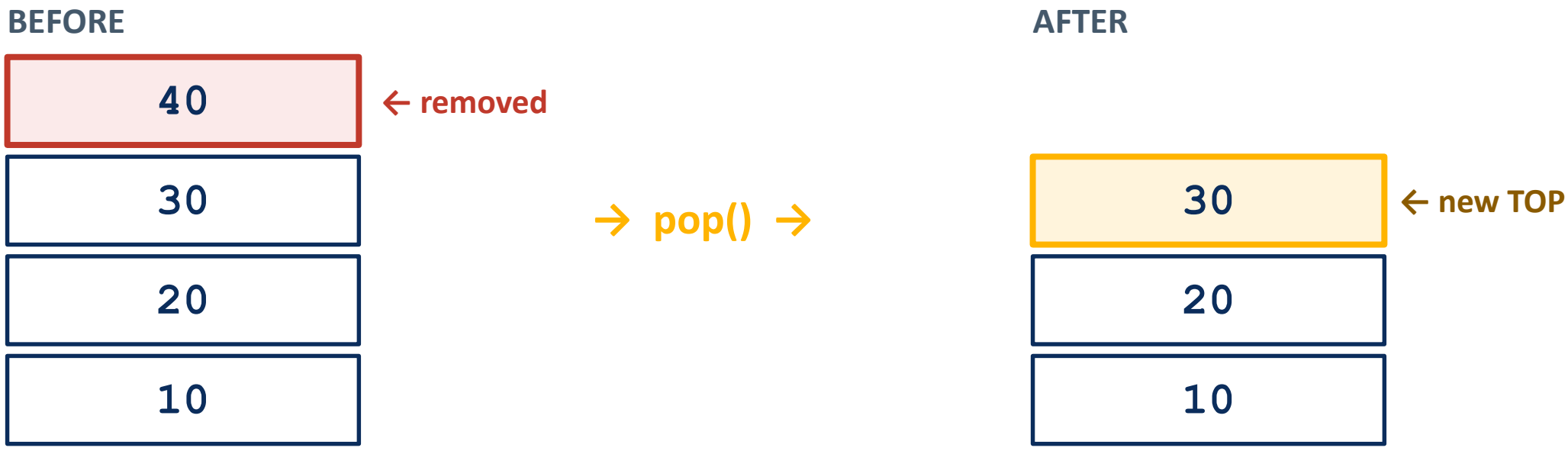
Stack Operation: Push

Push adds a new element to the TOP of the stack.



Stack Operation: Pop

Pop removes the TOP element from the stack and returns it.



Peek, isEmpty & isFull

1. **Peek** — Look at the TOP element WITHOUT removing it
2. **isEmpty** — Checks if the stack has zero elements
3. **isFull** — Checks if the stack has reached its maximum size

Exam focus: explain the difference between peek and pop

Array-Based Implementation

```
int  stack[5];    int  top = -1;
```

push(value)

```
top = top + 1;  stack[top] = value;
```

pop()

```
value = stack[top];  top = top - 1;
```

The variable "top" always points to the index of the current top element — that's the heart of the stack.

Teaching tip: trace top's value on the board through a few pushes and pops

Real-World Applications of Stack

1 Function Calls

Programs use a "call stack" to track active function calls and recursion

2 Undo / Redo

Apps store your actions on a stack to undo the most recent one first

3 Expression Evaluation

Compilers use stacks to evaluate and convert math expressions

4 Browser History

The back button pops the most recently visited page

Exam focus: name 3 real-world uses of a stack

Advantages & Disadvantages

ADVANTAGES

✓ Simple and easy to implement

✓ Push and Pop are very fast — O(1)

✓ Manages memory automatically in function calls

DISADVANTAGES

✗ Fixed size in array-based stacks

✗ No random access — only the top is reachable

✗ Risk of overflow (full) or underflow (empty)

Exam focus: give 2 advantages and 2 disadvantages of a stack

QUICK REVISION

Exam-Ready Summary: Stack

Definition	A linear structure following LIFO — Last In, First Out
Operations	Push (add), Pop (remove), Peek (view top), isEmpty, isFull
Applications	Function calls, undo/redo, expression evaluation, browser history
Trade-off	Very fast push/pop, but only the top element is accessible

Tip: draw the plate-stack diagram on the board for every push/pop question.